CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Decomposition of vegetable matter into compost is an example of: 1

(a) a displacement reaction **(b)** a physical change
(c) an endothermic reaction **(d)** an exothermic reaction

2. Which of the following is a strong base? 1

(a) copper hydroxide **(b)** ammonium hydroxide
(c) magnesium hydroxide **(d)** sodium hydroxide

3. A student states: “Bases produce OH^- ions in water.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only in acids **(d)** Cannot be determined

4. The number of valence electrons in a magnesium atom is: 1

(a) 3 **(b)** 1 **(c)** 8 **(d)** 2

5. The most ductile metal is: 1

(a) mercury **(b)** lead **(c)** gold **(d)** iron

6. The general formula of alkynes is: 1

(a) C_nH_{2n+2} **(b)** C_nH_{2n-2} **(c)** C_nH_{2n} **(d)** C_nH_n

7. Vinegar is a dilute solution of: 1

(a) ethanol **(b)** methanoic acid
(c) ethanoic acid **(d)** hydrochloric acid

8. In desert plants, stomata open at: 1

(a) noon **(b)** morning only **(c)** never **(d)** night

9. Picking up a pencil is controlled by the: 1

(a) spinal cord **(b)** hypothalamus **(c)** cerebellum **(d)** medulla

10. For DNA copying, Student A says it produces exactly identical copies; Student B says it produces copies with slight variations. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

11. The cutting and ligation of vas deferens is called: 1

(a) condom **(b)** tubectomy **(c)** vasectomy **(d)** copper-T

12. In the F2 of a dihybrid cross of RRYY × rryy, the number of round green plants out of 16 is: 1

(a) 3 **(b)** 1 **(c)** 6 **(d)** 9

13. The term 'gene' refers to a section of DNA that: 1

(a) provides information for one protein **(b)** stores fat

(c) makes energy **(d)** digests food

14. For a convex lens, Student A says its focal length is positive; Student B says negative. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

15. The defect of vision in which the near point is beyond 25 cm is: 1

(a) myopia **(b)** colour blindness **(c)** hypermetropia **(d)** astigmatism

16. The device used to change the resistance in a circuit is: 1

(a) ammeter **(b)** rheostat **(c)** fuse **(d)** voltmeter

17. Resistance of a 2 m wire is 4 Ω. A 6 m wire of the same material and thickness has resistance: 1

(a) 4/3 Ω **(b)** 12 Ω **(c)** 24 Ω **(d)** 8 Ω

18. The magnetic effect of electric current was discovered by: 1

(a) Oersted **(b)** Ohm **(c)** Newton **(d)** Faraday

19. The component which converts complex organic substances into simple inorganic ones is: 1

(a) decomposers **(b)** carnivores **(c)** herbivores **(d)** producers

20. Which waste can be converted into compost? 1

(a) metal cans **(b)** glass **(c)** plastic bottles **(d)** kitchen waste

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why does aluminium not corrode readily even though it is a reactive metal? 2

22. Why is a dilute acid a weaker acid than a concentrated acid of the same kind? 2

23. What are the differences between autotrophic and heterotrophic nutrition? 2

24. Why does a person suffering from diabetes need insulin injections? 2

25. What are the functions of the ovary in human females? 2

26. A student says acquired traits like muscle building are inherited. Justify or refute. 2

27. Name the type of mirror used in 2

(i) headlights

(ii) side mirrors of vehicles.

28. Draw (describe) the field lines around a straight wire carrying current upward. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Explain with equations what happens when 3

(i) zinc is added to copper sulphate solution

(ii) sodium sulphate is added to barium chloride solution

(iii) magnesium is burnt in air.

30. How is aluminium extracted? Write the reactions at the cathode and anode. 3

31. Explain how plant growth is influenced by hormones (auxin, gibberellin, cytokinin). 3

32. Explain the role of tissue culture in horticulture. 3

33. What is the function of the optic nerve and retina? What are rods and cones? 3

34. Explain the role of the earth wire in metal-bodied appliances. 3

35. Why does distilled water not conduct electricity whereas rain water does? 3

OR

Describe an activity to show that all acids contain hydrogen ions but not all hydrogen-containing compounds are acids. 3

36. What are the effects of depletion of the ozone layer? (three) 3

OR

Why is the number of organisms at the top of the food chain less? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A car's side mirror reads “Objects in mirror are closer than they appear”. 5

(i) Name the mirror.
(ii) Why is this warning written?
(iii) If $R = 2$ m, find f.
(iv) For a car 10 m behind, find the image distance.
(v) Is the image real?

OR

(i) Describe image formation by a concave mirror when the object is at 5
(a) C
(b) between C and F
(c) at F.
(ii) Draw (describe) any one ray diagram.

38. (i) Explain the structure of a soap molecule. 5
(ii) How does it remove oily dirt?
(iii) Why does soap not lather well in hard water?
(iv) Give one advantage of detergents.

OR

(i) Differentiate between ethanol and ethanoic acid on the basis of 5
(a) litmus
(b) $\mathrm{NaHCO_3}$
(c) smell.
(ii) Write the structures of both.

39. (i) How many resistors of 10 Ω each are needed to carry 2.5 A on a 5 V line when connected in parallel? 5
(ii) Show the combination of four 4 Ω resistors giving 4 Ω.

OR

MASTERMIND

(i) What is an electric circuit? Draw (describe) the symbols of a cell, battery, key, resistor, rheostat, ammeter and voltmeter. 5

(ii) Why should a voltmeter have a high resistance?

40. (i) Describe transportation of water and minerals in plants. 5

(ii) What is root pressure?

(iii) When is it important?

OR

(i) What is photosynthesis? Write its equation. 5

(ii) Explain the role of stomata.

(iii) What happens to excess glucose?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND